

JupyterLab interface showing a code cell execution. The code imports necessary libraries, connects to a MySQL database, and reads data from a table named 'sales'.

```
[1]: import mysql.connector
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.linear_model import LinearRegression

# MySQL Connection
conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="shamana@123",
    database="retail_dss"
)

query = """
SELECT sale_date, total_amount
FROM sales
ORDER BY sale_date
"""

df = pd.read_sql(query, conn)
print(df)
```

	sale_date	total_amount
0	2025-01-10	55000.0
1	2025-02-15	20000.0
2	2025-03-12	20000.0

C:\Users\SHAMANA\AppData\Local\Temp\ipykernel_11676\1879107638.py:21: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.

```
[2]: df['sale_date'] = pd.to_datetime(df['sale_date'])
df['month'] = df['sale_date'].dt.month

monthly_sales = df.groupby('month')['total_amount'].sum()
print(monthly_sales)

monthly_sales.plot(title="Monthly Sales Trend")
plt.show()
```

```
month
1    55000.0
2    20000.0
3    20000.0
Name: total_amount, dtype: float64
```



```
[6]: X = range(len(monthly_sales))
y = monthly_sales.values

model = LinearRegression()
model.fit([[i] for i in X], y)

future_months = [[len(X)], [len(X)+1]]
prediction = model.predict(future_months)

print("Predicted Sales:", prediction)
if prediction[-1] > monthly_sales.mean():
    print("Recommendation: Increase inventory and marketing spend.")
else:
    print("Recommendation: Offer discounts and improve customerengagement.")
```

```
Predicted Sales: [ -3333.33333333 -20833.33333333]
Recommendation: Offer discounts and improve customerengagement.
```

[]: